

Computer Programming

Lab 06

For Loop and Nested For Loop

1. Objective

The objectives of this lab are:

- To learn and apply the for loop.
- To learn and apply the nested for loop.
- To understand repetition and pattern printing in C++.

2. Introduction

A for loop is used when we know how many times a loop should run. It contains initialization, condition, and update statements. The syntax of a for loop is written as: for(initialization; condition; update). A nested for loop means a loop inside another loop. It is commonly used for tables, patterns, and matrix operations.

Task 1: Display Numbers from 1 to 10

Problem Statement

Write a C++ program that uses a loop to display numbers from 1 to 10. Each number should be printed on a separate line.

Program

```
#include <iostream>
using namespace std;

int main()
{
    // for loop starts from 1 and ends at 10
    for (int i = 1; i <= 10; i++)
    {
        // Display value of i
        cout << i << endl;
    }

    return 0;
}
```

Output

1
2
3
4
5
6
7
8
9
10

Task 2: Multiplication Table using Nested for Loop

Problem Statement

Write a C++ program that generates a multiplication table from 1 to 10 using nested for loops.

Program

```
#include <iostream>
using namespace std;

int main()
{
    // Outer loop for rows
    for (int i = 1; i <= 10; i++)
    {
        // Inner loop for columns
        for (int j = 1; j <= 10; j++)
        {
            // Print multiplication result
            cout << i * j << "\t";
        }

        // Move to next line after each row
        cout << endl;
    }

    return 0;
}
```

Output

```
1 2 3 4 5 6 7 8 9 10
2 4 6 8 10 12 14 16 18 20
3 6 9 12 15 18 21 24 27 30
```

Task 3: Sum of Numbers using for Loop

Problem Statement

Write a C++ program that asks the user to enter a positive integer. Use a for loop to calculate the sum from 1 up to the entered number.

Program

```
#include <iostream>
using namespace std;

int main()
{
    int num, sum = 0;

    // Take input from user
    cout << "Enter a positive number: ";
    cin >> num;

    // Calculate sum using for loop
    for (int i = 1; i <= num; i++)
    {
        sum = sum + i;
    }

    // Display result
    cout << "Sum = " << sum << endl;

    return 0;
}
```

Sample Output

```
Enter a positive number: 5
Sum = 15
```

Task 4: Triangle Pattern using Nested for Loop

Problem Statement

Write a C++ program that prints a triangle pattern of asterisks using nested for loops. The number of rows should be entered by the user.

Program

```
#include <iostream>
using namespace std;

int main()
{
    int rows;

    // Input number of rows
```

```

    cout << "Enter number of rows: ";
    cin >> rows;

    // Outer loop for rows
    for (int i = 1; i <= rows; i++)
    {
        // Inner loop for stars
        for (int j = 1; j <= i; j++)
        {
            cout << "* ";
        }

        // Move to next line
        cout << endl;
    }

    return 0;
}

```

Sample Output

```

*
**
***
****
*****

```

4. Conclusion

In this lab, we learned how to use for loops and nested loops in C++. We also learned how repetition works in programming, how to create multiplication tables and patterns, and how to calculate sums using loops.